Examen TP

Avant de répondre aux questions de TP,

1. Créer un utilisateur ayant votre nom (exemple : pour l'étudiant mohamed, useradd mohamed)
2. Ajoutez l'utilisateur mohamed au groupe wheel : (usermod –aG wheel mohamed)
3. Connectez vous en tant que mohamed (su mohamed)

Questions

1. Ajoutez quatre disques à votre machine (la taille de chaque disque est égale à 10G).

```
[alaa@localhost ~]$ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda      8:0    0   60G  0 disk
├─sda1   8:1    0  300M  0 part /boot
├─sda2   8:2    0    4G  0 part [SWAP]
└─sda3   8:3    0 55.8G  0 part /
sdb      8:16   0   10G  0 disk
sdc      8:32   0   10G  0 disk
sdd      8:48   0   10G  0 disk
sde      8:64   0   10G  0 disk
sr0     11:0    1 1024M  0 rom
```

2. Créez une partition à partir de disque /dev/sdb dont la taille est égale à 1G, configurez cette partition avec le système de fichiers ext4 et le point de montage /rep. Le montage doit être temporaire.

```
Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-20971519, default 2048):
Last sector, +sectors or +size{K,M,G,T,P} (2048-20971519, default 20971519): +1G

Created a new partition 1 of type 'Linux' and of size 1 GiB.
```

```
[alaa@localhost ~]$ sudo mkfs.ext4 /dev/sdb1
mke2fs 1.45.6 (20-Mar-2020)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: ad5405d2-f951-4a57-9b70-67e6208ab86e
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

```
[alaa@localhost ~]$ sudo mount /dev/sdb1 /home/alaa/rep/
[alaa@localhost ~]$ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda      8:0    0   60G  0 disk
├─sda1   8:1    0  300M  0 part /boot
├─sda2   8:2    0    4G  0 part [SWAP]
└─sda3   8:3    0 55.8G  0 part /
sdb      8:16   0   10G  0 disk
└─sdb1   8:17   0    1G  0 part /home/alaa/rep
sdc      8:32   0   10G  0 disk
sdd      8:48   0   10G  0 disk
sde      8:64   0   10G  0 disk
sr0     11:0    1 1024M  0 rom
```

3. Créez une partition swap de 1G à partir de disque /dev/sdc. Cette partition doit être montée lors du démarrage de manière automatique.

```
Command (m for help): n
Partition type
   p   primary (1 primary, 0 extended, 3 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (2099200-20971519, default 2099200):
Last sector, +sectors or +size{K,M,G,T,P} (2099200-20971519, default 20971519): +1G

Created a new partition 2 of type 'Linux' and of size 1 GiB.
```

```
Command (m for help): t
Partition number (1,2, default 2): 2
Hex code (type L to list all codes): 82

Changed type of partition 'Linux' to 'Linux swap / Solaris'.
```

```
[alaa@localhost ~]$ sudo mkswap /dev/sdb2
Setting up swapspace version 1, size = 1024 MiB (1073737728 bytes)
no label, UUID=9eb9e1cf-36ef-48a0-aa75-0295418edc90
```

```
[alaa@localhost ~]$ sudo swapon /dev/sdb2
[alaa@localhost ~]$
```

```
UUID=89a5888b-7a19-4f98-86de-5633ed75c36c none                    swap    defaults        0 0
```

```
[alaa@localhost ~]$ sudo vi /etc/fstab
[alaa@localhost ~]$ mount -a
mount: only root can use "--all" option
[alaa@localhost ~]$ sudo mount -a
```

4. Créez un pool stratis nommé pool contenant le disque /dev/sdb.

```
[alaa@localhost ~]$ sudo stratis pool create pool  /dev/sdc
```

5. Augmentez la capacité de **pool** à l'aide d'une partition /dev/sdc2 de taille 3G

```
[alaa@localhost ~]$ sudo stratis pool add-data pool /dev/sdd
```

6. Créez un système de fichiers nommé **fs1** dans le pool **pool**. Montez ce système de fichiers sur /labstratisvol afin qu'il persiste lors des redémarrages.

```
[alaa@localhost ~]$ sudo stratis filesystem create pool fs1
```

```
UUID=eb984c3b-7b52-4bcc-bece-bb4ba20b85ef /labstratisvol        xfs     x-systemd.requires=stratisd.service  0 0
```

```
[alaa@localhost ~]$ sudo vi /etc/fstab
[alaa@localhost ~]$ sudo mount -a
[alaa@localhost ~]$ sudo vi /etc/fstab
```

7. Créez un volume de groupe vg à partir des volumes physiques /dev/sdd1 (taille 1G) et /dev/sdd2 (taille 3G). Créez un volume logique **lv0** formaté en ext4 et monté automatiquement sous /data, avec une taille 1G.

```
Command (m for help): n
Partition type
   p   primary (1 primary, 0 extended, 3 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (2099200-20971519, default 2099200):
Last sector, +sectors or +size{K,M,G,T,P} (2099200-20971519, default 20971519): +3G

Created a new partition 2 of type 'Linux' and of size 3 GiB.
```

```
[alaa@localhost ~]$ pvcreate /dev/sdd{1,2}
  WARNING: Running as a non-root user. Functionality may be unavailable.
  /run/lock/lvm/P_global:aux: open failed: Permission denied
[alaa@localhost ~]$ sudo pvcreate /dev/sdd{1,2}
WARNING: sun signature detected on /dev/sdd1 at offset 508. Wipe it? [y/n]: y
  Wiping sun signature on /dev/sdd1.
  Physical volume "/dev/sdd1" successfully created.
  Physical volume "/dev/sdd2" successfully created.
```

```
[alaa@localhost ~]$ sudo vgcreate vg /dev/sde{1,2}
  WARNING: Not using device /dev/sdd1 for PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3.
  WARNING: PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3 prefers device /dev/sdd because device was seen first.
  Volume group "vg" successfully created
```

```
[alaa@localhost ~]$ sudo vgcreate vg /dev/sde{1,2}
  WARNING: Not using device /dev/sdd1 for PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3.
  WARNING: PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3 prefers device /dev/sdd because device was seen first.
  Volume group "vg" successfully created
```

```
[alaa@localhost ~]$ sudo lvdisplay
  WARNING: Not using device /dev/sdd1 for PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3.
  WARNING: PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3 prefers device /dev/sdd because device was seen first.
  --- Logical volume ---
  LV Path                /dev/vg/lv
  LV Name                lv
  VG Name                vg
  LV UUID                QmBUUb-7ZeR-6P8o-eS9f-N2mG-9IqI-YCHTmZ
  LV Write Access        read/write
  LV Creation host, time localhost.localdomain, 2021-06-25 12:32:15 -0700
  LV Status              available
  # open                 0
  LV Size                1.00 GiB
  Current LE             256
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     8192
  Block device           253:0
```

```
[alaa@localhost ~]$ mkfs.xfs /dev/vg/lv
mkfs.xfs: cannot open /dev/vg/lv: Permission denied
[alaa@localhost ~]$ sudo mkfs.xfs /dev/vg/lv
meta-data=/dev/vg/lv             isize=512    agcount=4, agsize=65536 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1
data     =                       bsize=4096   blocks=262144, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=2560, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
```

```
UUID=0073d3e3-3a8f-4f46-baa5-97781c358858 /data                   xfs     defaults        0 0
```

Mount -a

8. Changez la capacité du volume logique nommé l**vo** de 1G à 2G.

```
[alaa@localhost ~]$ sudo lvextend –r –L +1G /dev/vg/lv
  Logical volume name "–r" is invalid.
  Run `lvextend --help' for more information.
[alaa@localhost ~]$ sudo lvextend -r -L +1G /dev/vg/lv
  WARNING: Not using device /dev/sdd1 for PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3.
  WARNING: PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3 prefers device /dev/sdd because device was seen first.
Phase 1 - find and verify superblock...
Phase 2 - using internal log
        - zero log...
        - scan filesystem freespace and inode maps...
        - found root inode chunk
Phase 3 - for each AG...
        - scan (but don't clear) agi unlinked lists...
        - process known inodes and perform inode discovery...
        - agno = 0
        - agno = 1
        - agno = 2
        - agno = 3
        - process newly discovered inodes...
Phase 4 - check for duplicate blocks...
        - setting up duplicate extent list...
```

9. Est il possible de réduire la taille de volume logique **lv0** de 1G.

Oui (système de fichiers ext4)

10. Créez un volume logique **lv1** formaté en xfs et monté automatiquement sous /data2, avec une taille 500M.
11.

```
Command (m for help): n
Partition type
   p   primary (1 primary, 0 extended, 3 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (2099200-20971519, default 2099200):
Last sector, +sectors or +size{K,M,G,T,P} (2099200-20971519, default 20971519): +3G

Created a new partition 2 of type 'Linux' and of size 3 GiB.
```

12.

```
[alaa@localhost ~]$ pvcreate /dev/sdd{1,2}
  WARNING: Running as a non-root user. Functionality may be unavailable.
  /run/lock/lvm/P_global:aux: open failed: Permission denied
[alaa@localhost ~]$ sudo pvcreate /dev/sdd{1,2}
WARNING: sun signature detected on /dev/sdd1 at offset 508. Wipe it? [y/n]: y
  Wiping sun signature on /dev/sdd1.
  Physical volume "/dev/sdd1" successfully created.
  Physical volume "/dev/sdd2" successfully created.
```

13.

```
[alaa@localhost ~]$ sudo vgcreate vg /dev/sde{1,2}
  WARNING: Not using device /dev/sdd1 for PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3.
  WARNING: PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3 prefers device /dev/sdd because device was seen first.
  Volume group "vg" successfully created
```

14.

```
[alaa@localhost ~]$ sudo vgcreate vg /dev/sde{1,2}
  WARNING: Not using device /dev/sdd1 for PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3.
  WARNING: PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3 prefers device /dev/sdd because device was seen first.
  Volume group "vg" successfully created
```

15.

```
[alaa@localhost ~]$ sudo lvdisplay
  WARNING: Not using device /dev/sdd1 for PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3.
  WARNING: PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3 prefers device /dev/sdd because device was seen first.
  --- Logical volume ---
  LV Path                /dev/vg/lv
  LV Name                lv
  VG Name                vg
  LV UUID                QmBUUb-7ZeR-6P8o-eS9f-N2mG-9IqI-YCHTmZ
  LV Write Access        read/write
  LV Creation host, time localhost.localdomain, 2021-06-25 12:32:15 -0700
  LV Status              available
  # open                 0
  LV Size                1.00 GiB
  Current LE             256
  Segments               1
  Allocation             inherit
  Read ahead sectors     auto
  - currently set to     8192
  Block device           253:0
```

16.

```
[alaa@localhost ~]$ mkfs.xfs /dev/vg/lv
mkfs.xfs: cannot open /dev/vg/lv: Permission denied
[alaa@localhost ~]$ sudo mkfs.xfs /dev/vg/lv
meta-data=/dev/vg/lv             isize=512    agcount=4, agsize=65536 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1
data     =                       bsize=4096   blocks=262144, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=2560, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
```

17.

```
UUID=0073d3e3-3a8f-4f46-baa5-97781c358858 /data                   xfs     defaults        0 0
```

18. Changez la capacité du volume logique nommé l**v1** de 500M à 5G. (Vous avez la liberté de créer un nouveau volume physique pour répondre à cette question).

```
[alaa@localhost ~]$ sudo lvextend –r –L +1G /dev/vg/lv
  Logical volume name "–r" is invalid.
  Run `lvextend --help' for more information.
[alaa@localhost ~]$ sudo lvextend -r -L +1G /dev/vg/lv
  WARNING: Not using device /dev/sdd1 for PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3.
  WARNING: PV vk4XL3-IV0N-13bV-QpNa-MAQt-p1c7-gWimu3 prefers device /dev/sdd because device was seen first.
Phase 1 - find and verify superblock...
Phase 2 - using internal log
        - zero log...
        - scan filesystem freespace and inode maps...
        - found root inode chunk
Phase 3 - for each AG...
        - scan (but don't clear) agi unlinked lists...
        - process known inodes and perform inode discovery...
        - agno = 0
        - agno = 1
        - agno = 2
        - agno = 3
        - process newly discovered inodes...
Phase 4 - check for duplicate blocks...
        - setting up duplicate extent list...
```

19. Créez un volume vdo à partir de disque /dev/sde afin d'offrir une taille logique de 100G. Ce volume sera configuré avec le système de fichiers xfs et le point de montage /vdo1.

```
[alaa@localhost ~]$ sudo vdo create --name vdo2  --device /dev/sdc --vdoLogicalSize 100G
Creating VDO vdo2
      The VDO volume can address 6 GB in 3 data slabs, each 2 GB.
      It can grow to address at most 16 TB of physical storage in 8192 slabs.
      If a larger maximum size might be needed, use bigger slabs.
Starting VDO vdo2
Starting compression on VDO vdo2
VDO instance 0 volume is ready at /dev/mapper/vdo2
```

```
[alaa@localhost ~]$ sudo mkfs.xfs -K  /dev/mapper/vdo2
meta-data=/dev/mapper/vdo2       isize=512    agcount=4, agsize=6553600 blks
         =                       sectsz=4096  attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1
data     =                       bsize=4096   blocks=26214400, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=12800, version=2
         =                       sectsz=4096  sunit=1 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
```

```
/dev/mapper/vdo2                        /vdo1                xfs     x-systemd.requires=vdo.service,discard  0 0
```

20. Planifier une tache qui permet de compresser /tmp dans le fichier /home/tekup/tmp.tar.bz2 le 26 juin 2022.

```
[alaa@localhost ~]$ at june 26 22
warning: commands will be executed using /bin/sh
at> tar –cvfj /home/tekup/tmp.tar.bz2 /tmp
at> <EOT>
job 5 at Sun Jun 26 12:54:00 2022
```

21. Ecrire un script shell qui compte le nombre de fichiers dans un répertoire donné.

```
fonction calcul
{
 {
  repertoire='ls -l "$nom" '
  echo "calcul de $nom :"
  echo "le nbr de repertoire est de " 'echo "$repertoire" | grep -c "^d"'
  echo "le nbr de fichier executable est de " 'echo "$repertoire | grep -c "^[+]..x"'
  echo "le nbr total de fichier est de " 'echo "$repertoire" | grep -c "^.*"'
 }
}

if [$# -ne 0]
then
    for nom in $*
    do
      calcul $nom
    done
else
    echo "entrer un non de repertoire"
    read nom
```

```
    calcul $nom
fi
```